

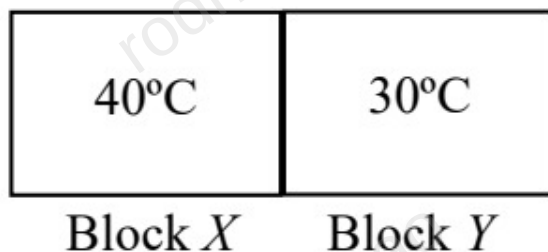
2012 HKDSE Physics

Paper 1A Multiple Choice Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1

Two metal blocks X and Y of the same mass and of initial temperatures 40°C and 30°C respectively are in good thermal contact as shown. The specific heat capacity of X is greater than that of Y. Which statement is correct when a steady state is reached? Assume no heat loss to the surroundings.



- A. The temperature of block X is higher than that of block Y.
- B. Their temperature becomes the same and is lower than 35°C .
- C. Their temperature becomes the same and is higher than 35°C .**
- D. Their temperature becomes the same and is equal to 35°C .

Official answer: C

Thermal equilibrium requires the same final temperature T . With no heat loss, $mc_X(40-T) = mc_Y(T-30)$.

Hence $T = (40c_X + 30c_Y)/(c_X + c_Y)$. Since $c_X > c_Y$, $T - 35 = 5(c_X - c_Y)/(c_X + c_Y) > 0$. Answer: C.

Question 2

When a patient's arm is wiped by a piece of cotton soaked with alcohol, the wiped area will feel cool as that patch of alcohol on the skin evaporates. Which statement explains this phenomenon?

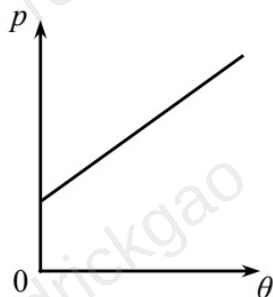
- A. The evaporation of alcohol absorbs heat from the patient's arm.**
- B. The alcohol on the skin releases latent heat to the surrounding air.
- C. The motion of all the molecules in the patch of alcohol slows down.
- D. Air molecules remove heat from the patch of alcohol by conduction.

Official answer: A

Evaporation requires latent heat, supplied in part by the warmer skin. The skin loses energy and cools: A. Faster molecules preferentially escape; this does not mean every remaining molecule slows down.

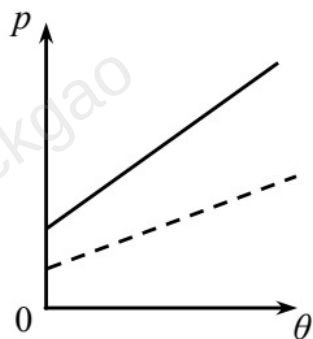
Question 3

An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ .

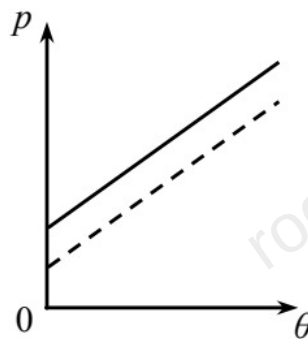


If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?

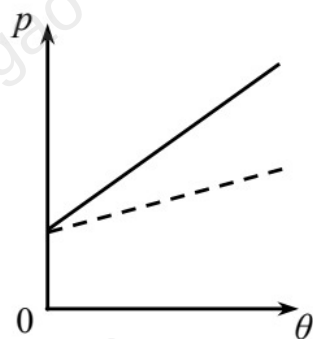
A.



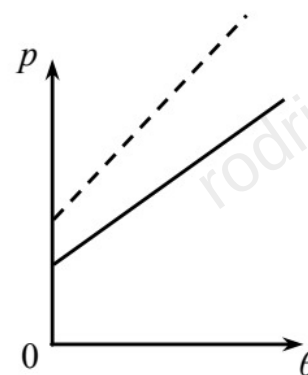
B.



C.



D.

**Official answer: A**

At fixed volume, $p = Nk(\theta + 273.15)/V$. Halving N halves the pressure at every Celsius temperature, so both the gradient and the vertical intercept halve. This is A; the extrapolated temperature-axis intercept remains -273.15°C .

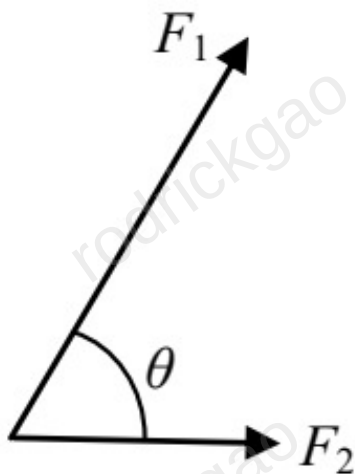
Question 4

Which of the following descriptions is correct?

- A. When water at 25°C is heated to 50°C, both the kinetic energy and potential energy of the water molecules increase.
- B. When water at 25°C is heated to 50°C, only the potential energy of the water molecules increases.
- C. When water boils at 100°C and turns into steam, the kinetic energy of the water molecules increases.
- D. When water boils at 100°C and turns into steam, the potential energy of the water molecules increases.

Official answer: D

During boiling at constant temperature, the average molecular kinetic energy does not increase. Latent heat increases molecular potential energy as molecules separate against attraction, so D is correct. In the syllabus model, heating without a change of state is associated with increased average molecular kinetic energy.

Question 5

Two forces F_1 and F_2 of constant magnitudes act at the same point as shown. When the angle θ between F_1 and F_2 increases from 0° to 180° , the magnitude of the resultant force

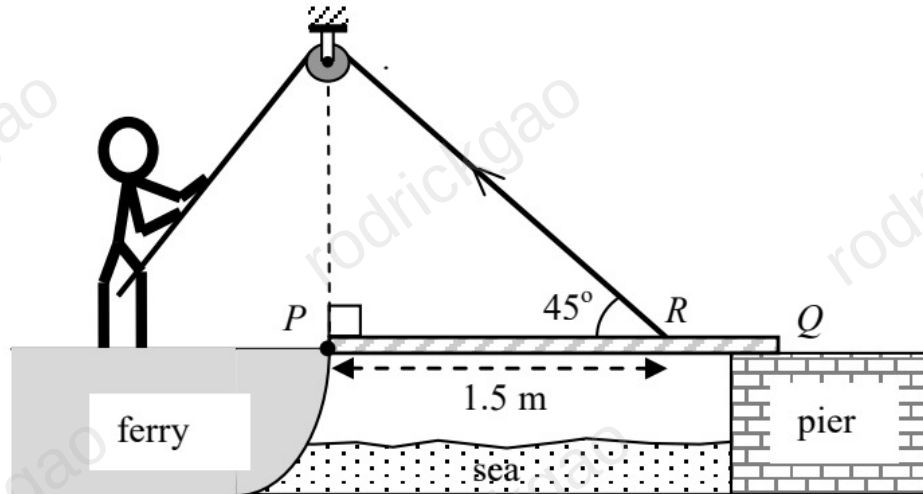
- A. decreases throughout.
- B. increases throughout.
- C. decreases and then increases.
- D. increases and then decreases.

Official answer: A

$R^2 = F_1^2 + F_2^2 + 2F_1F_2 \cos \theta$. As θ increases from 0° to 180° , $\cos \theta$ decreases from 1 to -1 . Therefore R decreases throughout, from $F_1 + F_2$ to $|F_1 - F_2|$: A.

Question 6

A uniform gangplank PQ of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m . It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P. Which of the following correctly describes the force required to raise the gangplank steadily?



- | | Initial force when horizontal | Subsequent force |
|----|-------------------------------|------------------------|
| A. | $0.67 Mg$ | greater than $0.67 Mg$ |
| B. | $0.67 Mg$ | smaller than $0.67 Mg$ |
| C. | $0.94 Mg$ | greater than $0.94 Mg$ |
| D. | $0.94 Mg$ | smaller than $0.94 Mg$ |

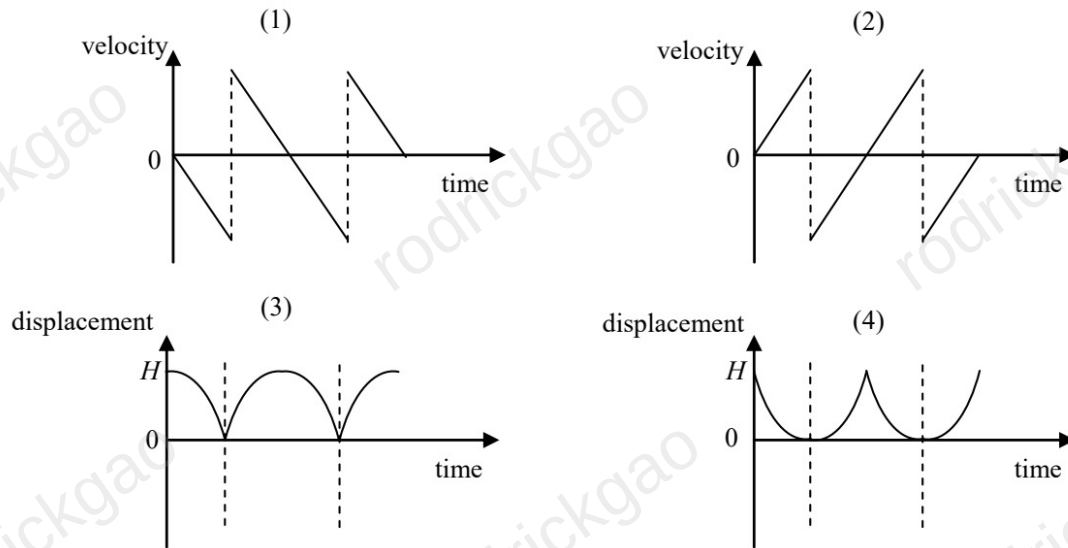
Official answer: D

Initially, moments about P give $T(1.5 \sin 45^\circ) = Mg(1)$, so $T = 0.943 Mg$. The light rope and smooth pulley make the man's pull equal to T .

To check the later force, let the plank rise through angle α , let $h = PR = 1.5\text{ m}$, and let d be the pulley-to-R rope length. Then $T(PR)h \cos \alpha / d = Mg(1) \cos \alpha$, so $T = Mg d / 2.25$ (with lengths in metres). As α increases, $d^2 = 1.5^2 + 1.5^2 - 2(1.5)^2 \sin \alpha$ decreases; hence T decreases. Answer: D.

Question 7

Which of the following graphs (velocity-time and displacement-time) best represent the motion of a ball falling from rest under gravity at a height H and bouncing back from the ground two times? Assume that the collision with the ground is perfectly elastic and neglect air resistance. (Downward measurement is taken to be negative.)



A. (1) and (3) only

B. (1) and (4) only

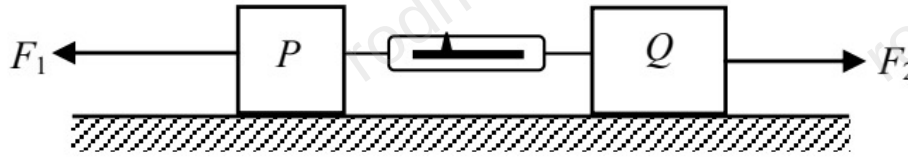
C. (2) and (3) only

D. (2) and (4) only

Official answer: A

Between impacts $a = -g$, so the velocity-time slope is negative. At each elastic bounce velocity changes from $-v$ to $+v$ without losing speed: graph (1).

The displacement is continuous, has negative curvature between impacts, and returns to height H . Its slope reverses abruptly at the ground but is zero at each highest point: graph (3). Answer: A.

Question 8

Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance?

- A. $(2F_1 - F_2)/3$
- B. $2(F_1 - F_2)/3$
- C. $(2F_1 + F_2)/3$
- D. $(F_1 + 2F_2)/3$

Official answer: C

For the whole system, $a = (F_1 - F_2)/(3m)$ to the left. For Q alone, $T - F_2 = 2ma$.

Therefore $T = F_2 + 2(F_1 - F_2)/3 = (2F_1 + F_2)/3$: C. The balance reads the tension, not the net force on the pair.

Question 9

An object of mass 0.5 kg is raised vertically from the ground by a motor. The object is raised 2.5 m in 1.5 s with uniform speed. Estimate the output power of the motor. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

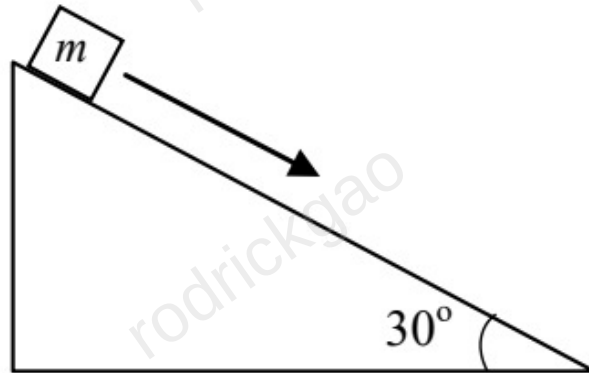
- A. 5.5 W
- B. 8.2 W
- C. 11.0 W
- D. 16.4 W

Official answer: B

Uniform speed means no kinetic-energy increase. Useful work is mgh , so $P = mgh/t = 0.5 \times 9.81 \times 2.5 / 1.5 = 8.175 \text{ W} \approx 8.2 \text{ W}$: B.

Question 10

A block of mass m resting on a 30° incline is given a slight push and slides down the incline with a uniform speed. Which of the following statements about the block's motion on the incline is/are correct?



- (1) There is no net force acting on the block.
- (2) The frictional force acting on the block is $0.5 mg$.
- (3) If the block is given a greater initial speed, it will slide down the incline with acceleration.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: C

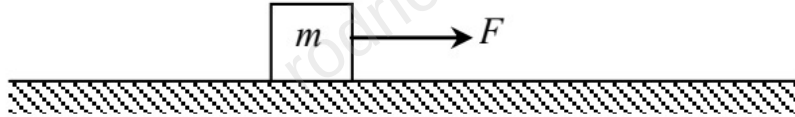
(1) Constant velocity gives $a = 0$ and hence zero resultant force.

(2) Along the slope, friction balances $mg \sin 30^\circ = 0.5 mg$.

(3) A greater initial speed does not change these forces in the stated friction model; acceleration remains zero.

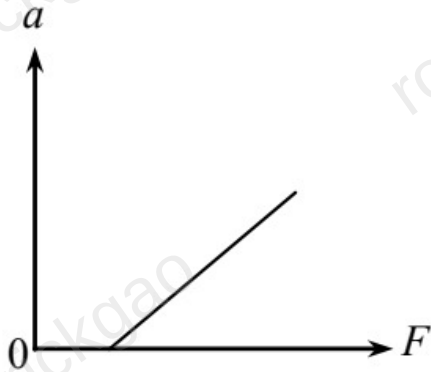
Only (1) and (2) are correct: C.

Question 11

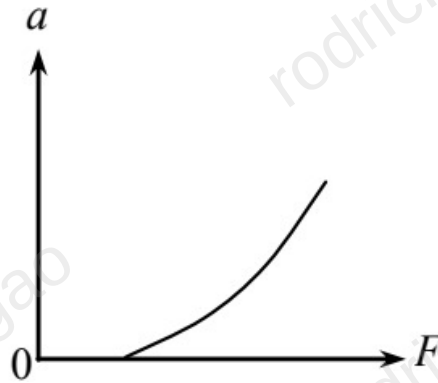


A block of mass m initially resting on a rough horizontal surface is pulled along the surface by a horizontal force F increasing from zero. If the frictional force is constant, which graph shows the relation between the acceleration of the block a and force F ?

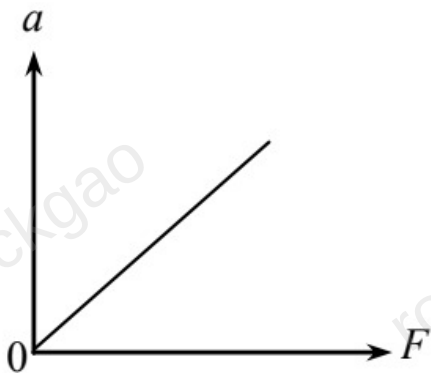
A.



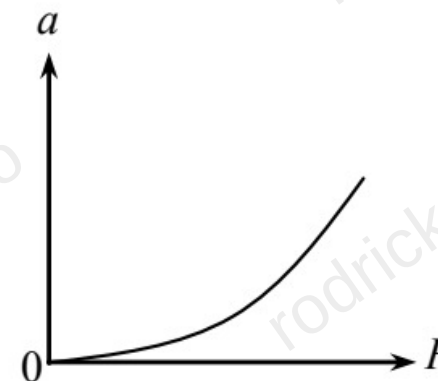
B.



C.



D.



Official answer: A

The block stays at rest until the pull overcomes friction. Once it slides against constant friction f , $a = (F-f)/m$, a straight line of gradient $1/m$ with a positive F -axis intercept. Thus A, not a curve or a line through the origin.

Question 12

A bomber aircraft is 1 km above the ground and is flying horizontally at a speed of 200 m s^{-1} . The aircraft is going to release a bomb to destroy a target on the ground. How long before flying over the target should the bomb be released? Assume that the bomber aircraft and the target are in the same vertical plane and neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

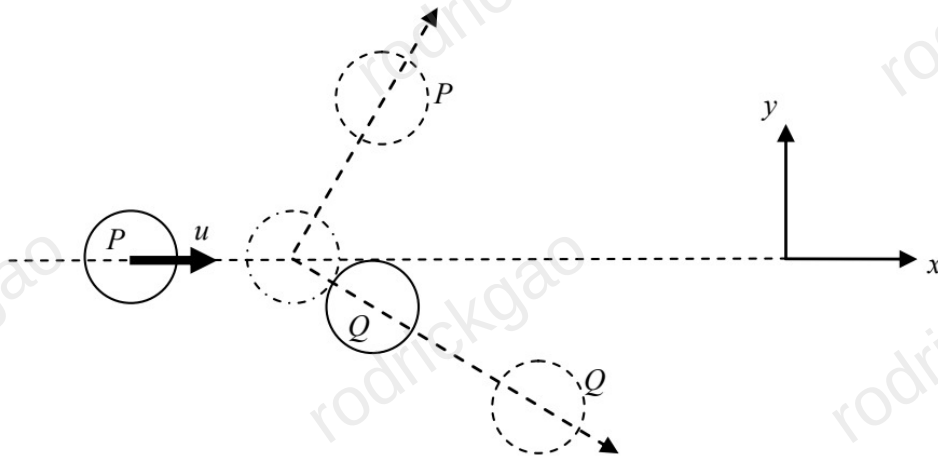
- A. 5.6 s
- B. 10.1 s
- C. 14.3 s
- D. It cannot be calculated as the horizontal distance between the aircraft and the target is not known.

Official answer: C

This is a vertical free-fall calculation: the initial vertical velocity is zero, so $1000 = \frac{1}{2}gt^2$ and $t = \sqrt{(2000/9.81)} = 14.3 \text{ s}$. With air resistance neglected, the released object retains the aircraft's horizontal velocity. The required lead time therefore equals its falling time: C.

Question 13

On a smooth horizontal surface, a circular disk P moving at velocity u along the x direction collides obliquely with an identical disk Q initially at rest as shown below. The mass of each disk is m . Which statements about the collision is/are correct?



- (1) Momentum of the system in the y direction is not conserved.
- (2) The total kinetic energy of P and Q after collision is $\frac{1}{2}mu^2$ if the collision is perfectly elastic.
- (3) Speed of Q after collision is less than u .

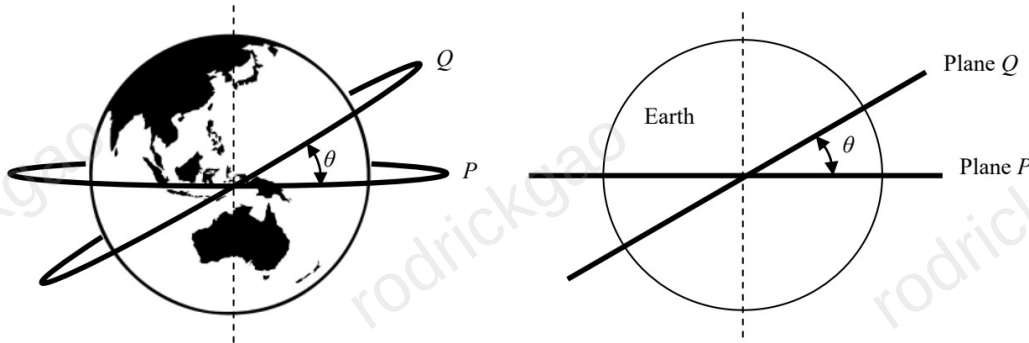
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

- (1) False: there is no external horizontal force, so momentum is conserved in both x and y ; total y -momentum remains zero.
- (2) True: a perfectly elastic collision conserves the initial total kinetic energy $\frac{1}{2}mu^2$.
- (3) True for the ordinary collision shown: total kinetic energy cannot increase, and P still moves after impact. Thus $\frac{1}{2}mv_Q^2$ is strictly less than $\frac{1}{2}mu^2$, giving $v_Q < u$. Answer: D.

Question 14

Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ . Which statement is INCORRECT?



- A. The speed of satellite P is $\sqrt{GM/R}$.
- B. The centripetal force acting on satellite Q is pointing along the plane Q.
- C. The acceleration of both satellites is the same in magnitude.
- D. The period of satellite Q is longer than that of satellite P.

Official answer: D

Gravity supplies centripetal force: $GMm/R^2 = mv^2/R$. Thus $v = \sqrt{GM/R}$, $a = GM/R^2$, and $T = 2\pi\sqrt{R^3/GM}$.

Both satellites have the same R , so the same speed, acceleration magnitude and period, regardless of orbital inclination. The force points towards Earth's centre within each orbital plane. D is the incorrect statement.

Question 15

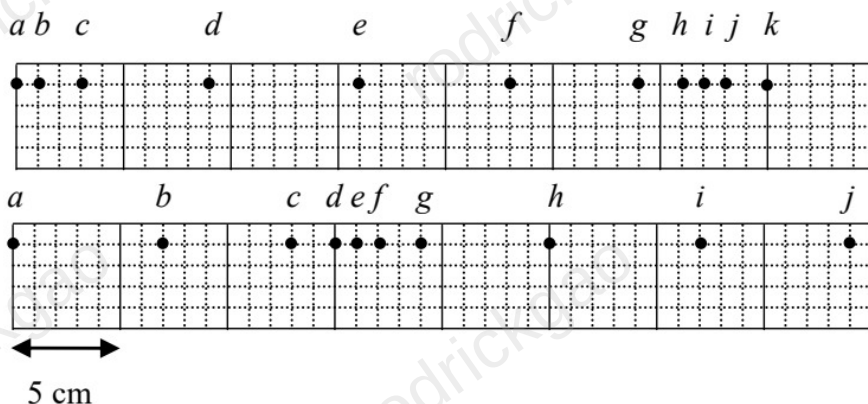


Figure (a)

Figure (b)

A series of particles is uniformly distributed along a slinky spring initially. Figure (a) shows their positions at a certain instant when a travelling wave propagates along the slinky spring from left to right. Figure (b) shows their positions 0.1 s later. Which statement is correct?

- A. Particle e is always stationary.
- B. Particles a and i are in phase.
- C. The wavelength of the wave is 16 cm.
- D. The frequency of the wave is 10 Hz.

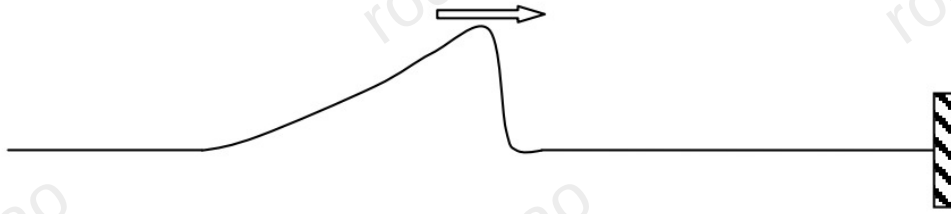
Official answer: B

Particles a and i occupy corresponding positions one complete spatial cycle apart, so they oscillate in phase: B.
The scale gives about 32 cm for this wavelength, not the 16 cm from a to e.

A travelling wave does not have a permanently stationary particle e merely because e is at the same position in these two snapshots. Also, 10 Hz would give a period of 0.1 s, requiring the whole pattern to repeat; the two patterns are different.

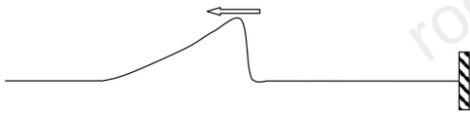
Question 16

A pulse on a string propagates towards the right end which is fixed.



Which of the following represents the reflected pulse?

A.



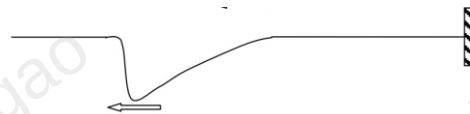
B.



C.



D.

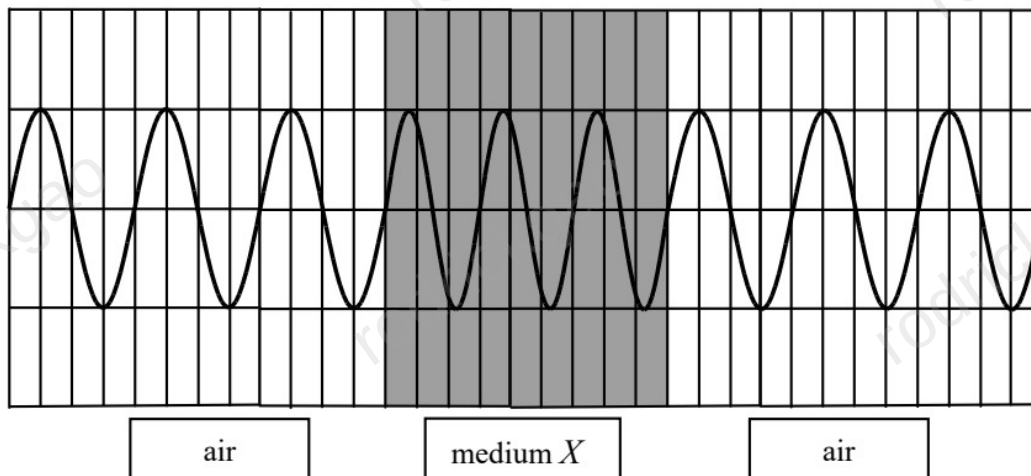


Official answer: D

Reflection at a fixed end reverses the displacement, so the reflected pulse is below the equilibrium line. It also travels left: the steep leading edge, which reaches the wall first, returns first and is now on the left. The shape is therefore inverted and horizontally reversed, as in D, not merely inverted as in C.

Question 17

A certain monochromatic light passes through medium X as shown below. What is the refractive index of medium X?



A. 1.25

B. 1.33

C. 1.50

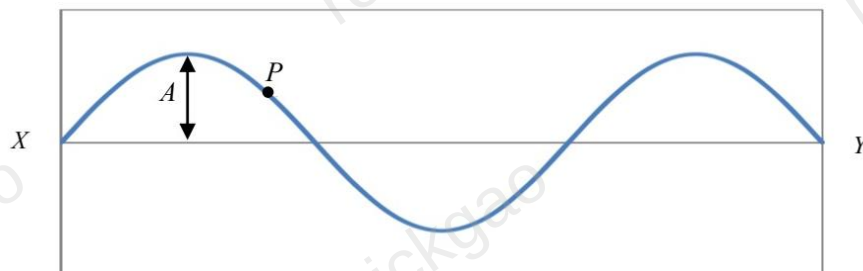
D. 1.65

Official answer: B

The frequency is unchanged at the boundary. From the grid, λ_{air} spans 4 divisions and λ_X spans 3. Hence $n_X \approx v_{\text{air}}/v_X = \lambda_{\text{air}}/\lambda_X = 4/3 = 1.33$: B.

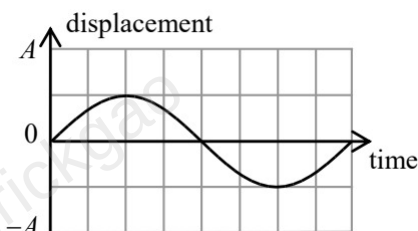
Question 18

A stationary wave is formed on a string fixed at both ends X and Y. The following is a snapshot of the string at time $t = 0$. The amplitude of vibration at an antinode is A.

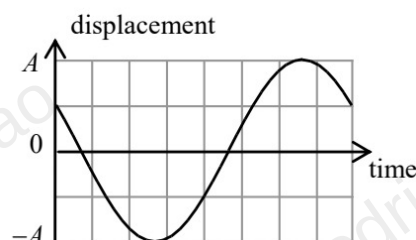


Which of the following shows the displacement-time graph of point P on the string for one period? (Upward displacement is taken as positive.)

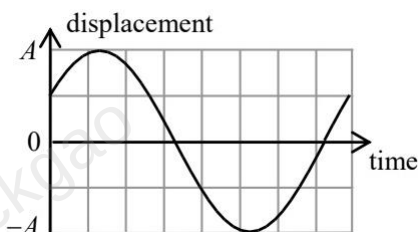
A.



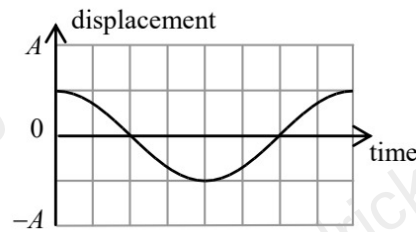
B.



C.



D.



Official answer: D

At $t = 0$ the antinode is at its maximum displacement A, so all non-node points are simultaneously at their own extreme displacements. P starts at its positive maximum, with zero instantaneous velocity.

P is not an antinode: its amplitude is less than A. Only D both starts at this positive maximum and has the smaller amplitude.

Question 19

Which of the following statements is INCORRECT?

- A. In air, the wavelength of infra-red radiation is shorter than that of ultra-violet radiation.
- B. Visible light travels faster in air than in glass.
- C. Microwaves travel at the speed of light in a vacuum.
- D. Both light and sound exhibit diffraction.

Official answer: A

The wavelength order is infrared > visible > ultraviolet, so A reverses the correct relationship. B and C follow the propagation speeds of electromagnetic waves, while diffraction is a property of both light and sound.

Question 20

For a diffraction grating of 600 lines per mm, the diffracted red light (657 nm) coincides with the diffracted violet light (438 nm) at angle of diffraction 52° . What are the respective orders of the diffracted red light and violet light?

	red	violet
A.	2	3
B.	3	4
C.	3	2
D.	4	3

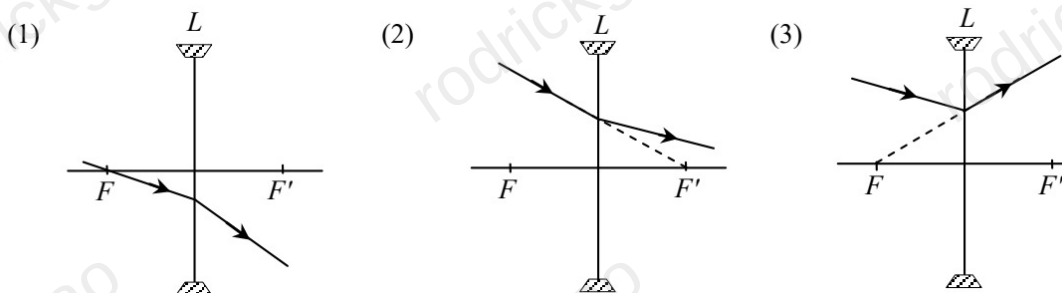
Official answer: A

The slit spacing is $d = 10^{-3}/600 = 1.667 \times 10^{-6}$ m. Using $d \sin \theta = n\lambda$ gives $n_{\text{red}} \approx 2.00$ and $n_{\text{violet}} \approx 3.00$: A.

Check: $2 \times 657 = 3 \times 438 = 1314$ nm, so the two orders have the same path difference and diffraction angle.

Question 21

In each of the following diagrams, L is a concave lens and its two principal foci are denoted by F and F'. Which of the ray diagrams is/are possible?



- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

Official answer: A

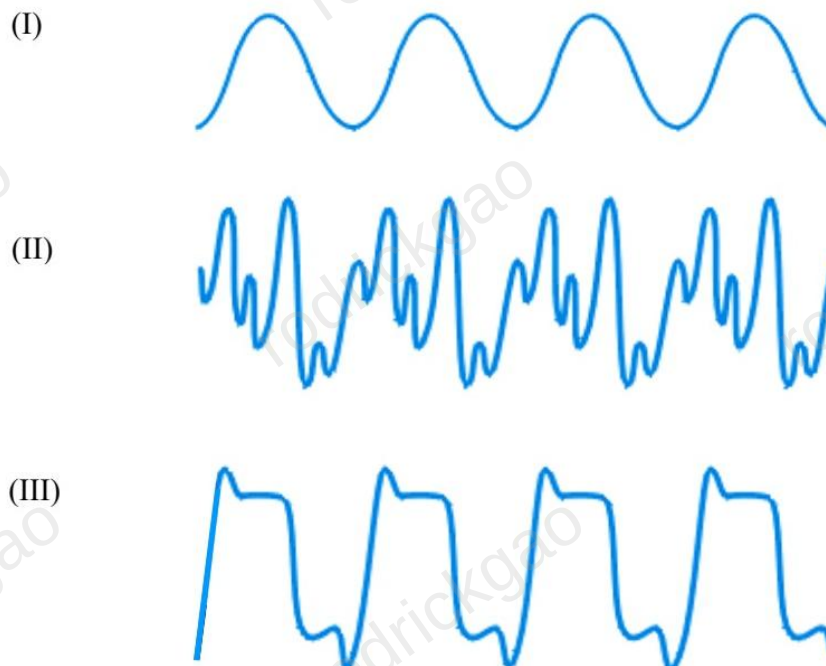
(1) is possible: below the principal axis, a concave lens bends the ray farther away from that axis. A ray through the near focus is not required to emerge parallel for a concave lens.

(2) is impossible: a ray aimed at the far focus F' must emerge parallel to the axis.

(3) is impossible: an emergent ray whose backward extension passes through the near focus F must have an incident ray parallel to the axis, which is not shown. Answer: A.

Question 22

The figure shows the waveforms of sound notes generated by a violin, a piano and a tuning fork. The scale is the same in time and intensity axes for all three waveforms.



Which of the following about the sound notes are correct?

- (1) They all have the same pitch.
- (2) The qualities of sound of (II) and (III) are different.
- (3) (I) is generated by the tuning fork.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

- (1) Each complete pattern repeats after the same time, so the fundamental frequency and pitch are the same. Do not count every small peak as a separate cycle.
 - (2) Different shapes mean different harmonic content and hence different sound quality (timbre).
 - (3) The nearly sinusoidal waveform (I) corresponds to the tuning fork's nearly pure tone. All three are correct.
- D.

Question 23

Which of the following about ultrasound is INCORRECT?

- A. Ultrasound is a longitudinal wave.
- B. The frequency of ultrasound is greater than 20000 Hz.
- C. In air, the speed of ultrasound is faster than the speed of audible sound.
- D. In air, the diffraction effect of ultrasound is less prominent than that of audible sound.

Official answer: C

In the same air under the same conditions, ultrasound and audible sound have approximately the same speed; higher frequency does not imply higher speed. Hence C is incorrect.

Since $\lambda = v/f$, ultrasound has a shorter wavelength, giving less diffraction for the same obstacle or aperture. Its frequency is above 20 kHz and its propagation in air is longitudinal.

Question 24

P, Q, R, S are charged objects. When two of them are brought close to each other, P and Q repel, R and S also repel while Q and R attract each other. Which of the following descriptions about their charges is/are possible?

- (1) P and R are negatively charged.
- (2) Q and S are positively charged.
- (3) P is positively charged and S is negatively charged.

A. (1) only

B. (3) only

C. (1) and (2) only

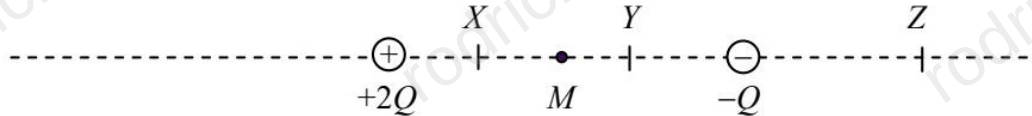
D. (2) and (3) only

Official answer: B

Repulsion makes P and Q have the same sign, and R and S have the same sign. Since all four are charged and Q attracts R, the two pairs have opposite signs.

Thus P and R cannot both be negative, and Q and S cannot both be positive. P positive with S negative is possible: only (3), answer B.

Question 25



Two point charges $+2Q$ and $-Q$ are situated at fixed positions as shown. M is the mid-point between the charges. X , Y and Z are points marked on the line joining these two charges. At which point could

(1) the resultant electric field due to the two charges be zero?

(2) the total electric potential due to the two charges be zero?

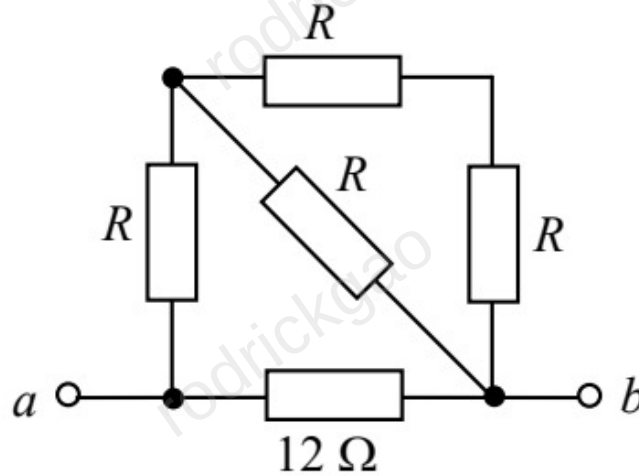
	(1)	(2)
A.	Z	X
B.	Z	Y
C.	X	Z
D.	Y	Z

Official answer: B

(1) Between opposite charges, both fields point from $+2Q$ towards $-Q$ and cannot cancel. Outside, cancellation can occur only on the smaller charge's side, represented by Z .

(2) Potential is a signed scalar: $2kQ/r_+ - kQ/r_- = 0$ requires $r_+ = 2r_-$. Between the charges this is nearer $-Q$ than the midpoint, as at Y , not X . The matching pair is **B**.

Question 26



In the above network, the resistance across terminals a and b is $6\ \Omega$. If the $12\ \Omega$ resistor is replaced by a $6\ \Omega$ resistor, the resistance across terminals a and b

- A. becomes $2\ \Omega$.
- B. becomes $4\ \Omega$.
- C. becomes $6\ \Omega$.
- D. cannot be found as the value of R is unknown.

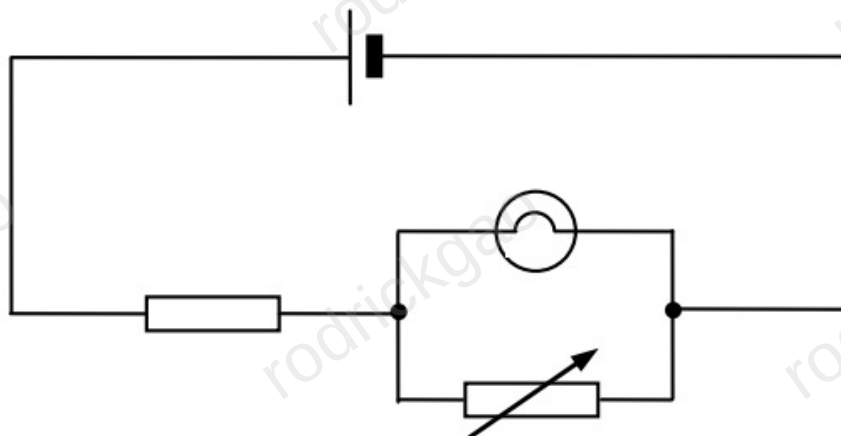
Official answer: B

Treat all four R resistors as one equivalent resistance S between a and b. Initially, S is in parallel with $12\ \Omega$: $\frac{1}{6} = \frac{1}{12} + \frac{1}{S}$, so $S = 12\ \Omega$.

After replacement, the equivalent is $6 \parallel 12 = \frac{(6 \times 12)}{(6 + 12)} = 4\ \Omega$: B. Finding the individual R is unnecessary.

Question 27

What will happen if the variable resistor is set to zero in the circuit below?

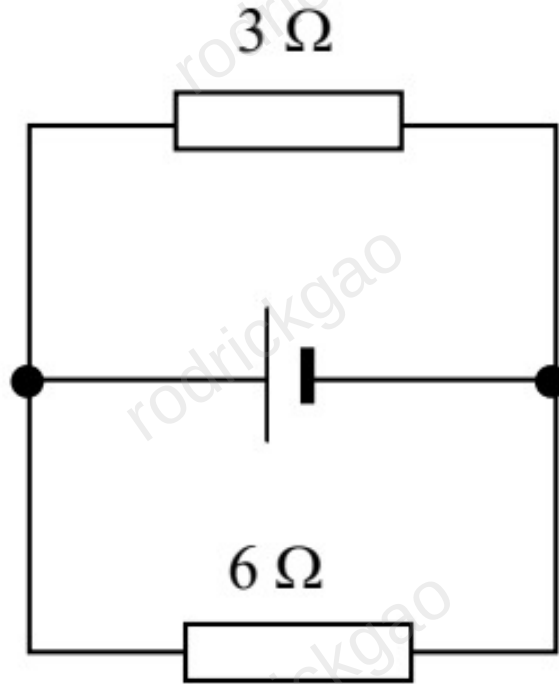


- A. The light bulb will burn out.
- B. The light bulb will not light up.**
- C. The brightness of the light bulb will increase.
- D. The brightness of the light bulb will remain unchanged.

Official answer: B

The zero-resistance branch short-circuits the lamp, making the potential difference across it zero. The lamp therefore carries no current and does not light: B. The separate series resistor still limits the circuit current.

Question 28



In the above circuit, the cell has e.m.f. 12 V and internal resistance $2\ \Omega$. What is the current in the $6\ \Omega$ resistor?

A. 0.5 A

B. 1.0 A

C. 1.5 A

D. 2.0 A

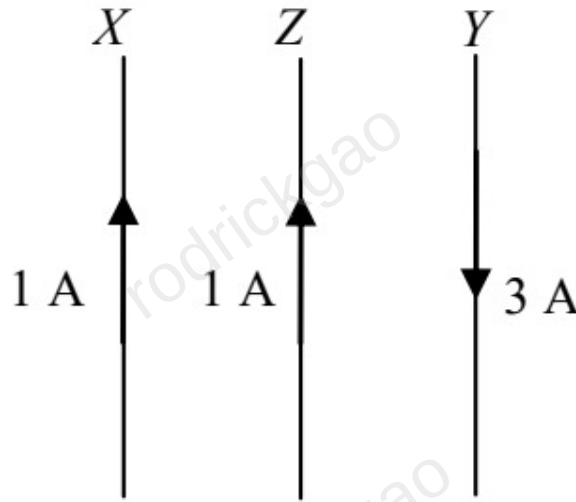
Official answer: B

The external resistance is $3\parallel 6 = 2\ \Omega$. Including the internal resistance gives $I_{\text{total}} = 12/(2+2) = 3\ \text{A}$.

Terminal voltage = $12 - 3 \times 2 = 6\ \text{V}$, so the $6\ \Omega$ branch current is $6/6 = 1.0\ \text{A}$: B. Do not apply the full e.m.f. directly across that branch.

Question 29

In the figure below, X, Y and Z are three long straight parallel wires with Z placed midway between X and Y. X and Z carry currents of 1 A in the same direction while Y carries a current of 3 A in the opposite direction. The magnetic force per unit length experienced by wire X due to wire Z is of magnitude F .



The magnetic force per unit length acting on wire Z due to both X and Y is

- A. $2F$ to the right.
- B. $2F$ to the left.
- C. $4F$ to the right.
- D. $4F$ to the left.

Official answer: D

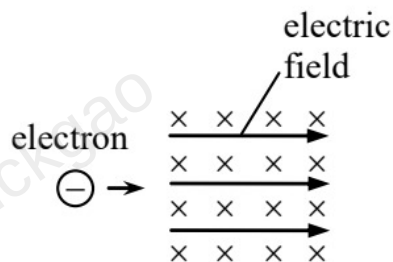
X and Z have parallel currents, so X attracts Z to the left with force per unit length F . Y has the opposite current and repels Z, also to the left.

For parallel wires, force per unit length is proportional to $I_1 I_2 / d$. The distances are equal and Y carries three times X's current, so its contribution is $3F$. Total = $4F$ left: D.

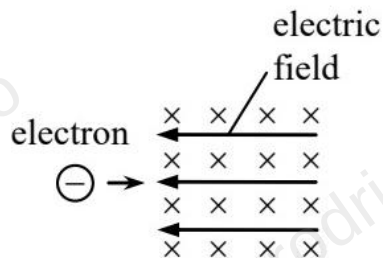
Question 30

An electron enters a region in which both a uniform electric field E and a uniform magnetic field B exist. The magnetic field B is pointing into the paper. In which direction should the electric field be applied so that the electron could be undeflected?

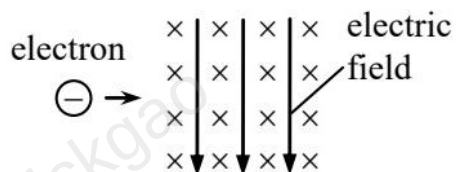
A.



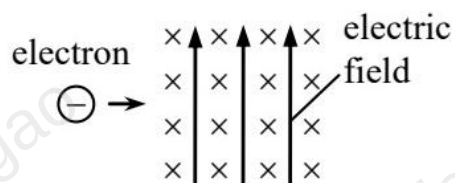
B.



C.



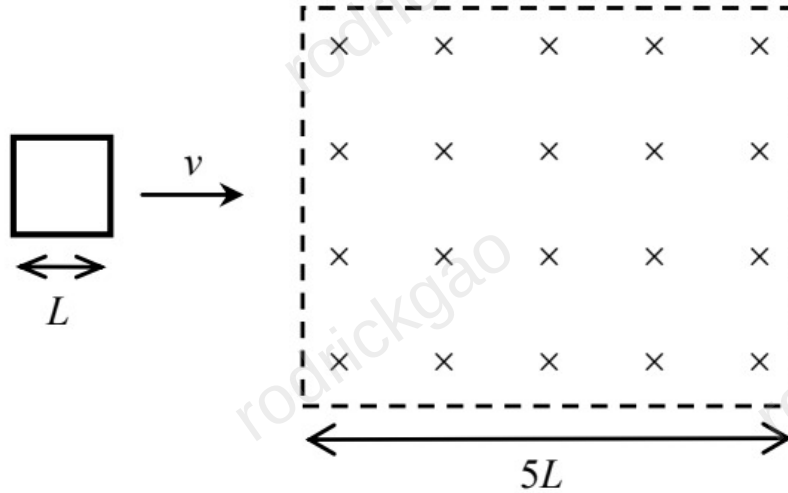
D.

**Official answer: C**

With velocity to the right and B into the page, the magnetic force on a negative electron is downward. The electric force must therefore be upward.

Because $F_E = qE$ and q is negative, the electric field must point downward: C. Its magnitude must also satisfy $E = vB$ for exact cancellation.

Question 31



A square metal frame of length of side L moving with constant velocity v passes through a region of uniform magnetic field of width $5L$ as shown. What is the total time period during which a current is induced in the frame?

- A. L/v
- B. $2L/v$
- C. $3L/v$
- D. $4L/v$

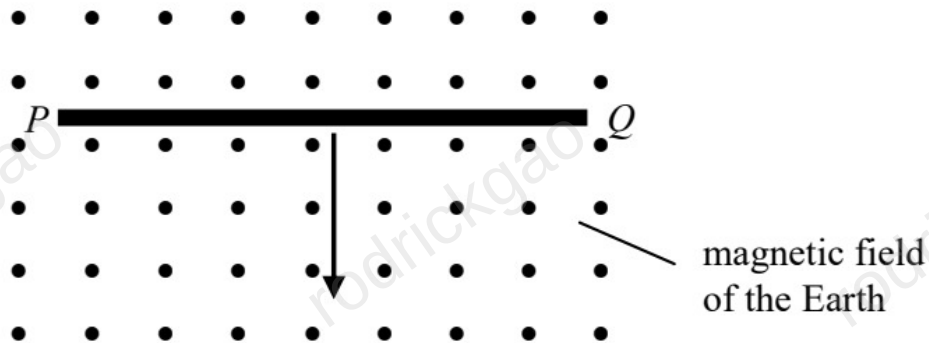
Official answer: B

Current is induced only while the magnetic flux through the frame changes. Entry takes L/v and exit takes another L/v .

While the entire frame is inside the uniform field, the flux is constant and there is no induced current. Total active time = $2L/v$: B, not the total crossing time.

Question 32

A copper rod PQ is placed horizontally as shown below. It is released and then falls vertically, cutting across the magnetic field of the Earth pointing out of the paper. Neglect air resistance. Which of the following statements is/are correct?



- (1) A voltage is induced across PQ.
- (2) A steady induced current is generated in the rod.
- (3) Due to the effect of the Earth's magnetic field, the copper rod falls with an acceleration less than the acceleration due to gravity.

A. (1) only

B. (3) only

C. (1) and (2) only

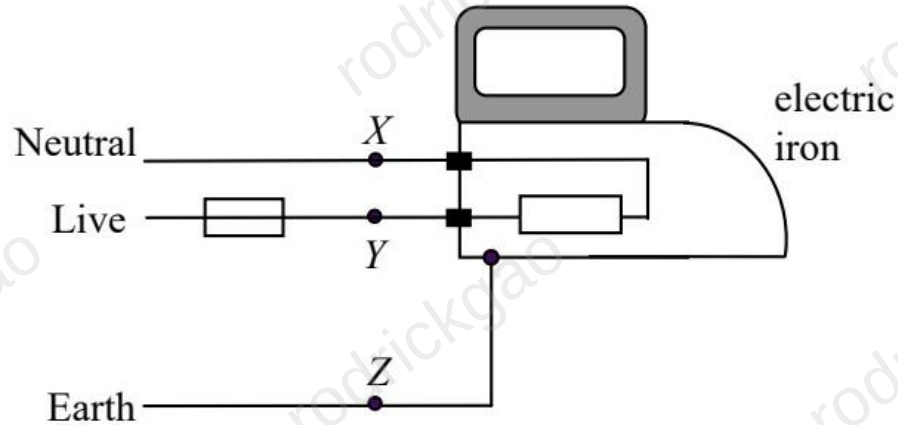
D. (2) and (3) only

Official answer: A

The magnetic force separates charges along the moving rod and produces a motional voltage $\varepsilon = BLv$. But the isolated rod is not a closed circuit, so no steady induced current flows.

Without such a current, there is no sustained magnetic braking force in this model; the rod falls with acceleration g . Only (1) is correct: A.

Question 33



The figure shows a simple domestic circuit for an electric iron. The fuse will blow when which of the following points are short-circuited?

- (1) X and Y
- (2) Y and Z
- (3) X and Z

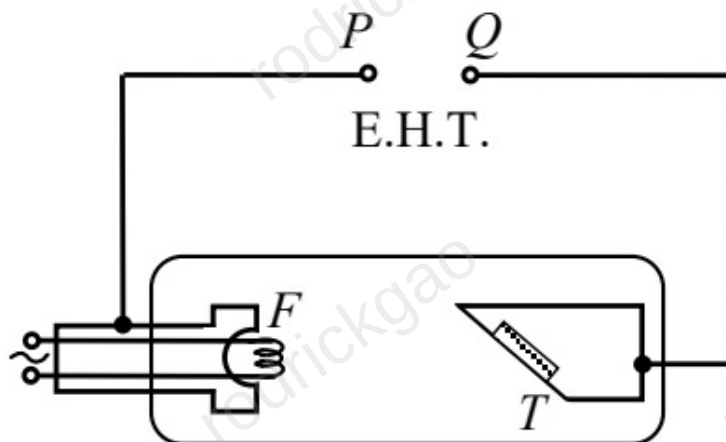
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

X is neutral, Y is live after the fuse, and Z is earth. An X–Y or Y–Z short provides a low-resistance path from live through the fuse, producing a large current that blows it.

X and Z are both approximately at earth potential in the ideal domestic-circuit model, so joining them does not create that large live current. Only (1) and (2): C. This is a paper-circuit analysis, not an experiment to perform.

Question 34



The figure shows a schematic diagram of an X-ray tube in which the filament F and the metal target T are connected to terminals P and Q of an E.H.T. Which statement is correct?

- A. P is the positive terminal and X-rays are emitted from T.
- B. P is the positive terminal and X-rays are emitted from F.
- C. Q is the positive terminal and X-rays are emitted from T.
- D. Q is the positive terminal and X-rays are emitted from F.

Official answer: C

The hot filament F emits electrons by thermionic emission. They must accelerate towards the positive target T, which is connected to Q. X-rays are produced when the electrons strike the target, not at the filament. Thus Q is positive and emission is from T: C.

Question 35

A certain radioactive isotope X has a half-life of 20 hours. After a time interval of 10 hours, what is the approximate fraction (f) of a sample of the radioactive isotope X remaining?

- A. $\frac{1}{4} \leq f \leq \frac{1}{2}$
- B. $f = \frac{1}{2}$
- C. $\frac{3}{4} > f > \frac{1}{2}$
- D. $f > \frac{3}{4}$

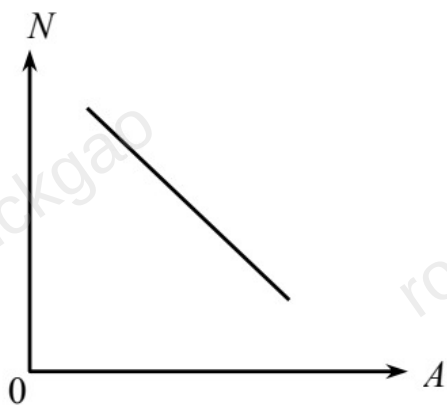
Official answer: C

Radioactive decay is exponential: $f = (\frac{1}{2})^{t/T_{1/2}} = (\frac{1}{2})^{10/20} = 1/\sqrt{2} \approx 0.707$. This lies strictly between 0.5 and 0.75: C.
Half a half-life does not mean a linear loss of one quarter.

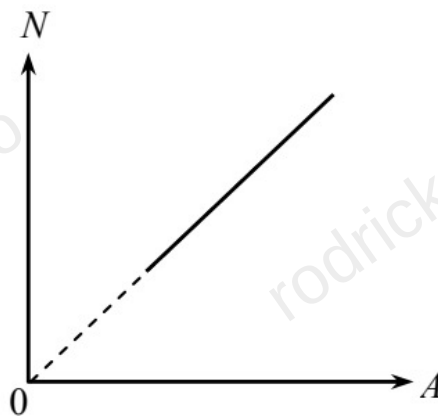
Question 36

Isotopes of an element have different mass number A and neutron number N . Which of the following N - A plots correctly shows the relationship of N and A for any given element?

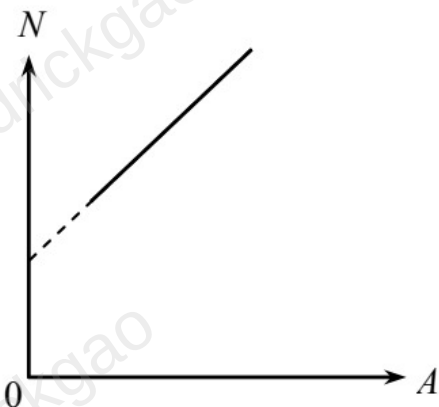
A.



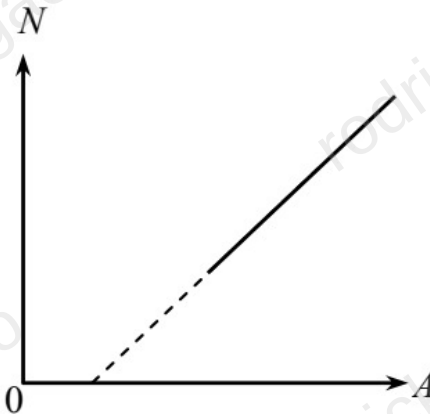
B.



C.



D.

**Official answer: D**

Isotopes of the same element have the same proton number Z . Since $A = Z + N$, $N = A - Z$.

The graph therefore has positive slope 1 and crosses the A -axis at $A = Z > 0$, not at the origin. This is D. The drawn line expresses the relationship; actual isotope values are integers.